159.302 LECTURE 12

Propositional Logic Syntax

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Reference: MIT OpenCourseWare

Propositional Logic

- **What is a logic?**

  A formal language

Propositional Logic

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  - **Syntax** – what expressions are legal

Propositional Logic

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  - **Syntax** – what expressions are legal
  - **Semantics** – what legal expressions mean

Propositional Logic

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A formal language

- **Syntax** – what expressions are legal
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- **Proof System** – a way of manipulating syntactic expressions to get other syntactic expressions (which will tell us something new)

Propositional Logic

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Why proofs?

- Two kinds of inferences an agent might want to make:

Propositional Logic

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  - **Multiple percepts** => conclusions about the world

Propositional Logic

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Why proofs?

- Two kinds of inferences an agent might want to make:
  - **Multiple percepts** => conclusions about the world
  - **Current state & operator** => properties of next state

Propositional Logic

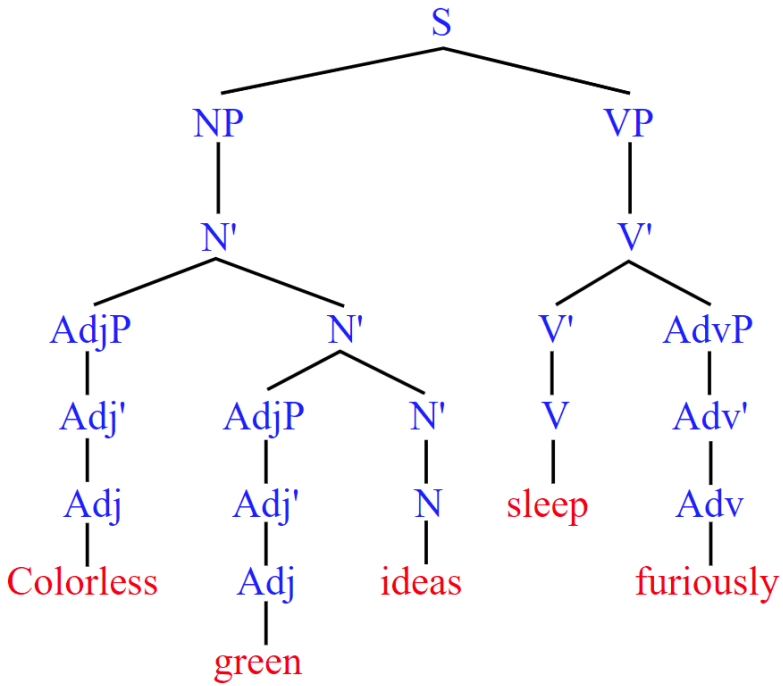
Syntax: what you're allowed to write

- for(thing t=fizz; t == fuzz; t++){…}

Propositional Logic

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Sentences (wffs: well-formed formulas)

Propositional Logic

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- Propositional variables are sentences: **P, Q, R, Z**

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- If **φ** and **ψ** are sentences, then so are

$$(\phi), \neg\phi, \phi \vee \psi, \phi \wedge \psi, \phi \rightarrow \psi, \phi \leftrightarrow \psi$$

Propositional Logic

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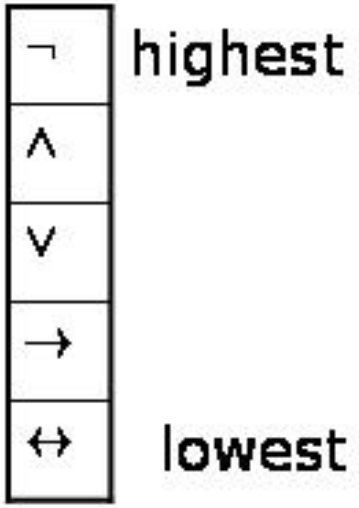
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- If **φ** and **ψ** are sentences, then so are

$$(\phi), \neg\phi, \phi \vee \psi, \phi \wedge \psi, \phi \rightarrow \psi, \phi \leftrightarrow \psi$$

- Nothing else is a sentence

Precedence



$A \vee B \wedge C$	$A \vee (B \wedge C)$
$A \wedge B \rightarrow C \vee D$	$(A \wedge B) \rightarrow (C \vee D)$
$A \rightarrow B \vee C \leftrightarrow D$	$(A \rightarrow (B \vee C)) \leftrightarrow D$

- Precedence rules enable "shorthand" form of sentences, but formally only the fully parenthesized form is legal.
- Syntactically ambiguous forms allowed in shorthand only when semantically equivalent: $A \wedge B \wedge C$ is equivalent to $(A \wedge B) \wedge C$ and $A \wedge (B \wedge C)$

159.302 LECTURE 2

Propositional Logic Semantics

Source: MIT OpenCourseWare

Propositional Logic

Semantics

- Meaning of a sentence is truth value {**t**, **f**}

Propositional Logic

Semantics

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holds(ϕ , i) [Sentence ϕ is **t** in interpretation i]

Propositional Logic

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Propositional Logic

Semantic Rules

holds(true, i) for all i

Propositional Logic

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Propositional Logic

Semantic Rules

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holds($\neg\phi$, i) if and only if fails(ϕ, i) **(negation)**

Propositional Logic

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holds($\phi \wedge \psi$, i) iff holds(ϕ,i) and holds(ψ, i) **(conjunction)**

Propositional Logic

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holds($\phi \vee \psi$, i) iff holds(ϕ,i) or holds(ψ, i) **(disjunction)**

holds(P, i) iff i(P) = **t**

fails(P, i) iff i(P) = **f**

Propositional Logic

Some important shorthand

$\phi \rightarrow \psi \equiv \neg\phi \vee \psi \quad (conditional, implication)$ **(antecedent $\rightarrow$ consequent)**

Propositional Logic

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P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$Q \rightarrow P$	$P \leftrightarrow Q$
f	f	t	f	f	t	t	t
f	t	t	f	t	t	f	f
t	f	f	f	t	f	t	f
t	t	f	t	t	t	t	t

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t	f	f	f	t	f	t	f
t	t	f	t	t	t	t	t

Note that implication is not “causality”, if P is **f** then $P \rightarrow Q$ is **t**

Implication doesn't mean "is related" in any kind of real-world way

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f	t	t	f	t	t	f	f
t	f	f	f	t	f	t	f
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Note that implication is not “causality”, if P is **f** then $P \rightarrow Q$ is **t**

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Propositional Logic

Terminology

A sentence is **valid** iff its truth value is **t** in all interpretations

Valid sentences : $\underline{true}$, $\neg\underline{false}$, $P \vee \neg P$

A sentence is **satisfiable** iff its truth value is **t** in at least one interpretation

Satisfiable sentences: $P, \underline{true}, \neg P$

A sentence is **unsatisfiable** iff its truth value is **f** in all interpretations

Unsatisfiable sentences : $P \wedge \neg P$, $\underline{false}$, $\neg\underline{true}$

All are finitely decidable.

Examples

Sentence	Valid?	Interpretation that make sentence's truth value = **f**
$smoke \rightarrow smoke$ $\neg smoke \vee smoke$	valid	

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Examples

Sentence	Valid?	Interpretation that make sentence's truth value = **f**
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contrapositive $(s \rightarrow f) \rightarrow (\neg f \rightarrow \neg s)$	valid	
$b \vee d \vee (b \rightarrow d)$ $b \vee d \vee \neg b \vee d$	valid	

Satisfiability

- Related to constraint satisfaction

- Given a sentence S, try to find an interpretation i such that ***holds(S,i)***

- Analogous to finding an assignment of values to variables such that the constraints hold

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Satisfiability

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- Given a sentence S, try to find an interpretation i such that ***holds(S,i)***
- Analogous to finding an assignment of values to variables such that the constraints hold
- Brute force method: enumerate all interpretations and check
- Better methods:

Heuristic search
Constraint propagation
Stochastic search

Satisfiability Problems

- Scheduling nurses to work in a hospital
  - propositional variables represent for example, that Pat is working on Tuesday at 2pm (= P).
  - constraints on the schedule are represented using logical expressions over the variables
- Finding bugs in software
  - propositional variables represent state of the program
  - use logic to describe how the program works and to assert there is a bug
  - if the sentence is satisfiable, you've found the bug!

159.302 LECTURE 3

Propositional Logic Proof

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Reference: MIT OpenCourseWare

Flowers will be pollinated?

- Imagine we knew that:

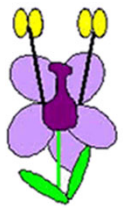
  - If today is Sunny, then the bees will be Foraging $(S \rightarrow F)$
  - If the bees are Foraging, the flowers will be Pollinated $(F \rightarrow P)$
  - Today is sunny (S)

Knowledge Base

Knowledge Base (KB)/Axioms/Premise/Givens/Facts

Should we conclude that the flowers will be pollinated? (P)





Checking Interpretations

S	F	P
t	t	t
t	t	f
t	f	t
t	f	f
f	t	t
f	t	f
f	f	t
f	f	f

8 possible interpretations

start by figuring out what set of interpretations make our original sentences true.

We can see that there are **8 possible interpretations**!

If **P** is true in a set of interpretations that make all the original sentences true, then we can conclude that **P** follows from our original sentences.

Checking Interpretations

interpretations *Knowledge base*

S	F	P	S→F	F→P	S
t	t	t	t	t	t
t	t	f	t	f	t
t	f	t	f	t	t
t	f	f	f	t	t
f	t	t	t	t	f
f	t	f	t	f	f
f	f	t	t	t	f
f	f	f	t	t	f

What set of interpretations make the KB true?

All 3 sentences (Knowledge Base) are true when S=t, F=t, and P=t.

What about the sentence P? Is it ok to conclude that it is also true?

Checking Interpretations

interpretations *sentences* *New sentence*

S	F	P	S→F	F→P	S	P
t	t	t	t	t	t	t
t	t	f	t	f	t	f
t	f	t	f	t	t	t
t	f	f	f	t	t	f
f	t	t	t	t	f	t
f	t	f	t	f	f	f
f	f	t	t	t	f	t
f	f	f	t	t	f	f

P is true in that interpretation!

If Knowledge Base is true, in the same set of interpretations that make P true (P is a new sentence we want to test), then it must be ok to conclude that P from the KB.



Adding a variable

- If Dandelions are blooming (D)

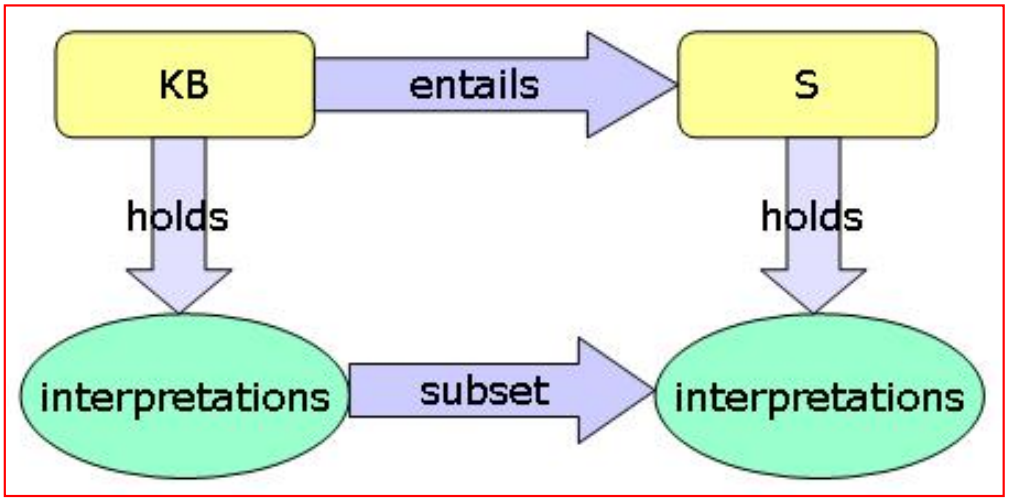
D	S	F	P	S → F	F → P	S	P
t	t	t	t	t	t	t	t
t	t	t	f	t	f	t	f
t	t	f	t	f	t	t	t
t	t	f	f	f	t	t	f
t	f	t	t	t	t	f	t
t	f	t	f	t	f	f	f
t	f	f	t	t	t	f	t
t	f	f	f	t	t	f	f
f	t	t	t	t	t	t	t
f	t	t	f	t	f	t	f
…	…	…	…				

If we add another variable (D), we can see that P is still true in two possible interpretations!



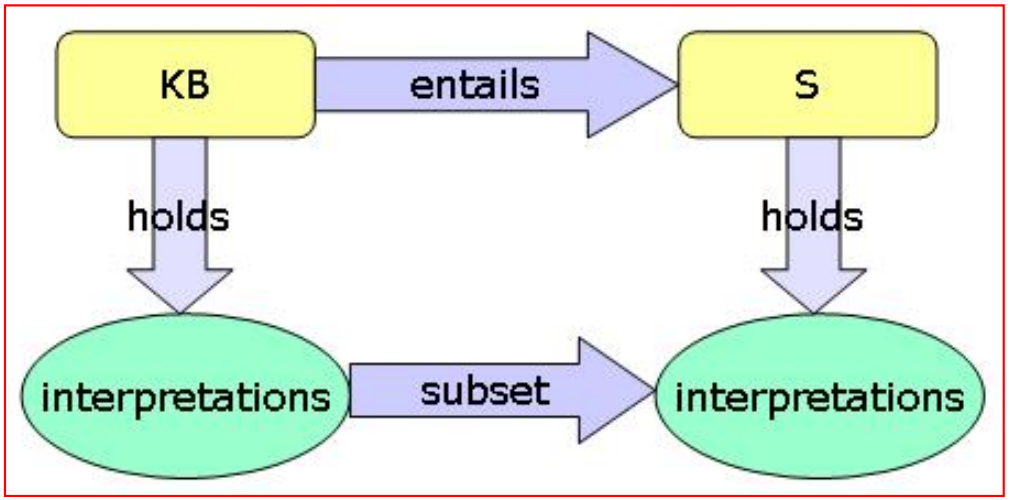
Entailment

•A knowledge base (**KB**) **entails** a sentence **S** iff every interpretation that makes **KB** true also makes **S** true



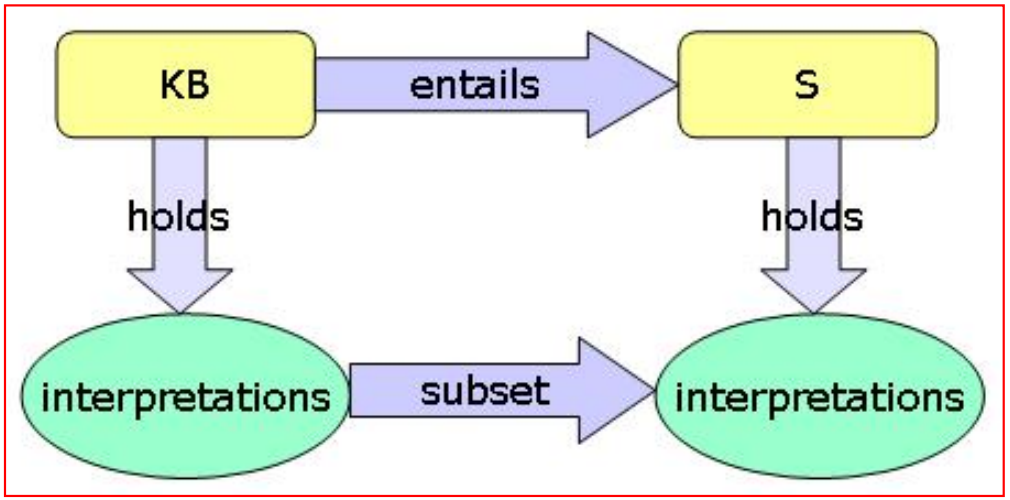
Computing Entailment

- enumerate all interpretations
- select those in which all elements of **KB** are true
- check to see if **S** is true in all of those interpretations



Computing Entailment

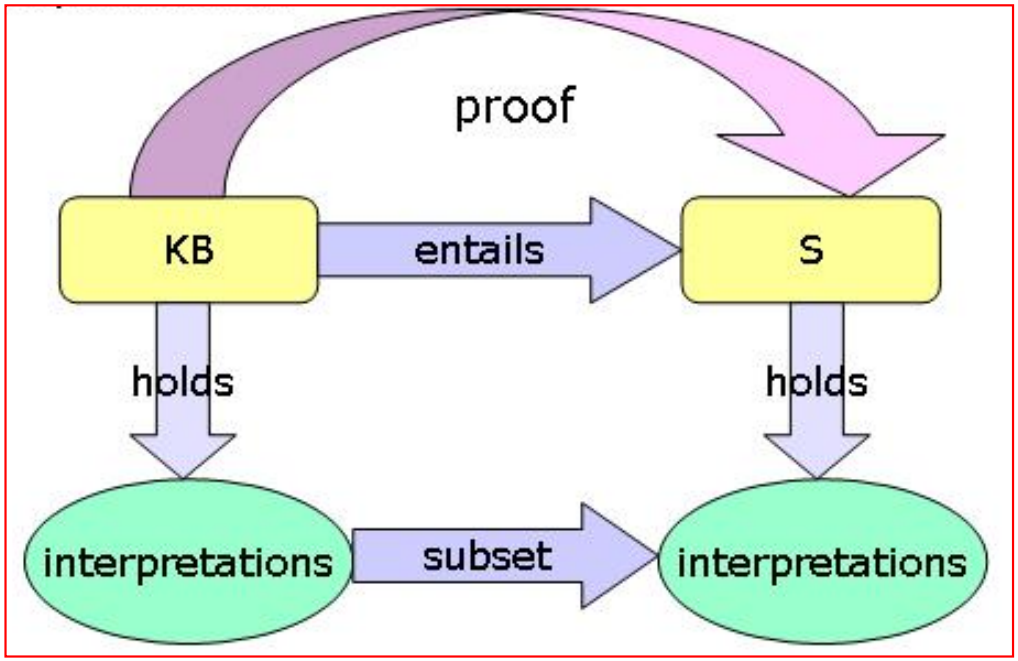
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- select those in which all elements of **KB** are true
- check to see if **S** is true in all of those interpretations



- This is brute force approach
- Way too many interpretations, in general!!
- e.g. (Given 100 propositional variables, there are 2^100 interpretations to check)

Entailment and Proof

• A **proof** is a way to test whether a **KB** entails a sentence, without enumerating all possible interpretations



Proof

- A **proof** is a sequence of sentences
- First ones are premises (**KB**)
- Then, you can write down on the next line the result of applying an inference rule to previous lines
- When **S** is on a line, you have **proved S** from **KB**
- If inference rules are **sound**, then any **S** you can prove from **KB** is entailed by **KB**
- If inference rules are **complete**, then any **S** that is entailed by **KB** can be proven from **KB**

Natural Deduction

- Some **inference rules**:

$$\frac{\alpha \rightarrow \beta \quad \alpha}{\beta}$$

Modus ponens

Affirming method

(first written by Aristotle)

Natural Deduction

- Some inference rules:

$$\frac{\alpha \to \beta \quad \alpha}{\beta}$$

Modus ponens

$$\frac{\alpha \to \beta \quad \neg\beta}{\neg\alpha}$$

Modus tolens

Denying method

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Modus ponens

$$\frac{\alpha \rightarrow \beta \quad \neg\beta}{\neg\alpha}$$

Modus tolens

$$\frac{\alpha \quad \beta}{\alpha \wedge \beta}$$

And-Introduction

Natural Deduction

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Modus ponens

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And-Introduction

$$\frac{\alpha \wedge \beta}{\alpha}$$

And-Elimination

Natural Deduction Example

Prove S

Step	Formula	Derivation
1	$P \wedge Q$	Given
2	$P \rightarrow R$	Given
3	$(Q \wedge R) \rightarrow S$	Given

Natural Deduction Example

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1	$P \wedge Q$	Given
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4	P	**1 And-Elim**

$$\frac{\alpha \wedge \beta}{\alpha}$$

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5	R	4,2 Modus-Ponens

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6	Q	1 And-Elim

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Natural Deduction Example

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6	Q	1 And-Elim
7	$Q \wedge R$	5,6 And-Intro

$$\frac{\alpha \quad \beta}{\alpha \wedge \beta}$$

Natural Deduction Example

Prove S

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1	$P \wedge Q$	Given
2	$P \rightarrow R$	Given
3	$(Q \wedge R) \rightarrow S$	Given
4	P	1 And-Elim
5	R	4,2 Modus-Ponens
6	Q	1 And-Elim
7	$Q \wedge R$	5,6 And-Intro
8	S	7,3 Modus-Ponens

$$\frac{\alpha \rightarrow \beta \quad \alpha}{\beta}$$

Proof Systems

• There are many natural deduction systems; they are typically "**proof checkers**", **sound** but not complete

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• **Natural deduction** uses lots of inference rules which introduces a **large branching factor** in the search for a proof.

• In general, you need to do "**proof by cases**" which introduces even more branching.

Prove R

Step	Formula
1	$P \vee Q$
2	$Q \rightarrow R$
3	$P \rightarrow R$

Propositional Resolution

Resolution rule:

$$\frac{\alpha \vee \beta \qquad \neg\beta \vee \gamma}{\alpha \vee \gamma}$$

- Single inference rule is a sound and complete proof system

- Requires all sentences to be converted to conjunctive normal form

159.302 LECTURE 4

Propositional Logic
Conjunctive Normal Form (CNF)

Source: MIT OpenCourseWare

Conjunctive Normal Form

CNF Formulas:

$$(A \vee B \vee \neg C) \wedge (B \vee D) \wedge (\neg A) \wedge (B \vee C)$$

Conjunctive Normal Form

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$(A \vee B \vee \neg C)$ is a clause

Conjunctive Normal Form

CNF Formulas:

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$A, B, and \neg C)$ are literals

Conjunctive Normal Form

CNF Formulas:

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- Every sentence in propositional logic can be written in CNF

Converting to CNF

1. Eliminate arrows using definitions



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2. Drive in negations using De Morgan's Laws

$$\neg(\phi \vee \varphi) \equiv \neg\phi \wedge \neg\varphi)$$

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Converting to CNF

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2. Drive in negations using De Morgan's Laws

$$\neg(\phi \vee \varphi) \equiv \neg\phi \wedge \neg\varphi)$$

$$\neg(\phi \wedge \varphi) \equiv \neg\phi \vee \neg\varphi)$$

3. Distribute **or** over **and**

$$(A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

Converting to CNF

1. Eliminate arrows using definitions

2. Drive in negations using De Morgan's Laws

$$\neg(\phi \vee \varphi) \equiv \neg\phi \wedge \neg\varphi)$$

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3. Distribute **or** over **and**

$$(A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

4. Every sentence can be converted to CNF, but it may **grow exponentially in size**

CNF Conversion Example

$$(A \vee B) \rightarrow (C \rightarrow D)$$

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$$(\neg A \vee \neg C \vee D) \wedge (\neg B \vee \neg C \vee D)$$

159.302 LECTURE 5

Propositional Resolution

Source: MIT OpenCourseWare

Propositional Resolution

Resolution rule:

$$\frac{\alpha \vee \beta \qquad \neg\beta \vee \gamma}{\alpha \vee \gamma}$$

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Resolution refutation:

- Convert all sentences to CNF, then write each clause down as a premise

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Resolution refutation:

- Convert all sentences to CNF, then write each clause down as a premise
- Negate the desired conclusion (converted to CNF)

This is proof by contradiction!

- Assert the sentence that you're trying to prove is false
- then try to derive a contradiction.

Propositional Resolution

Resolution rule:

$$\frac{\alpha \vee \beta \qquad \neg\beta \vee \gamma}{\alpha \vee \gamma}$$

Resolution refutation:

- Convert all sentences to CNF, then write each clause down as a premise
- Negate the desired conclusion (converted to CNF)
- Apply resolution rule until either
  - Derive false (a contradiction)
  - Can’t apply anymore

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Resolution refutation is sound and complete

- If we derive a contradiction, then the conclusion follows from the axioms
- If we can't apply anymore, then the conclusion cannot be proven from the axioms

Propositional Resolution

Prove R

Step	Formula
1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	**Negated conclusion**
5	$Q \vee R$	1,2
6	$\neg P$	2,4
7	$\neg Q$	3,4
8	R	5,7
9	▪	4,8

Proof steps

Small box indicates empty clause, meaning false

Propositional Resolution

Prove R

Step	Formula
1	$P \vee Q$
2	$P \rightarrow R$
3	$Q \rightarrow R$

$$\frac{\begin{array}{c} false \vee R \\ \neg R \vee false \end{array}}{false \vee false}$$

Step	Formula	Derivation
1	$P \vee Q$	Given
2	$\neg P \vee R$	Given
3	$\neg Q \vee R$	Given
4	$\neg R$	**Negated conclusion**
5	$Q \vee R$	**1,2**
6	$\neg P$	**2,4**
7	$\neg Q$	**3,4**
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Note: disjunction is commutative.

Propositional Resolution

Prove R

Step	Formula
1	$P \vee Q$
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Step	Formula	Derivation
1	$P \vee Q$	Given
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3	$\neg Q \vee R$	Given
4	$\neg R$	**Negated conclusion**
5	$Q \vee R$	1,2
6	$\neg P$	2,4
7	$\neg Q$	3,4
8	R	5,7
9	▪	4,8

unnecessary

$$\frac{false \vee R \qquad \neg R \vee false}{false \vee false}$$

The Power of False

Prove Z

Step	Formula
1	P
2	$\neg P$

Step	Formula	Derivation
1	P	Given
2	$\neg P$	Given
3	$\neg Z$	**Negated conclusion**
4	▪	1,2

Note that $(P \wedge \neg P) \rightarrow Z$ is valid

• Any conclusion follows from a contradiction – and so strict logic systems are very brittle.

Example Problem

Prove R

Step	Formula
1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

Convert to CNF

- ¬(¬ P ∨ Q) ∨ Q
- (P ∧ ¬ Q) ∨ Q
- (P ∨ Q) ∧ (¬ Q ∨ Q)
- (P ∨ Q)

- ¬(¬ P ∨ P) ∨ R
- (P ∧ ¬ P) ∨ R
- (P ∨ R) ∧ (¬ P ∨ R)

- ¬(¬ R ∨ S) ∨ ¬ (¬ S ∨ Q)
- (R ∧ ¬ S) ∨ (S ∧ ¬ Q)
- (R ∨ S) ∧ (¬ S ∨ S) ∧ (R ∨ ¬ Q) ∧ (¬ S ∨ ¬ Q)
- (R ∨ S) ∧ (R ∨ ¬ Q) ∧ (¬ S ∨ ¬ Q)

Example Problem

Prove R

Step	Formula
1	$(P \rightarrow Q) \rightarrow Q$
2	$(P \rightarrow P) \rightarrow R$
3	$(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

Step		
1	$P \vee Q$	
2	$P \vee R$	
3	$\neg P \vee R$	
4	$R \vee S$	
5	$R \vee \neg Q$	
6	$\neg S \vee \neg Q$	
7	$\neg R$	**Neg**

Example Problem

Prove R

Step	Formula
1	$(P \to Q) \to Q$
2	$(P \to P) \to R$
3	$(R \to S) \to \neg(S \to Q)$

Step		
1	$P \vee Q$	
2	$P \vee R$	
3	$\neg P \vee R$	
4	$R \vee S$	
5	$R \vee \neg Q$	
6	$\neg S \vee \neg Q$	
7	$\neg R$	Neg
8	S	4,7
9	$\neg Q$	6,8
10	P	1,9
11	R	3,10
12	■	7,11

Proof Strategies

Unit Preference: prefer a resolution step involving a unit clause (clause with one literal).

- Produces a shorter clause – which is good since we are trying to produce a zero-length clause, that is, a contradiction.

Set of support: choose a resolution involving the negated goal or any clause derived from the negated goal.

- We're trying to produce a contradiction that follows from the negated goal, so these are "relevant" clauses
- If a contradiction exists, one can find one using the set-of-support strategy.

References

MIT Opencourseware

Logic for Computer Science and Artificial Intelligence

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